Debugging Tool Quick Guide

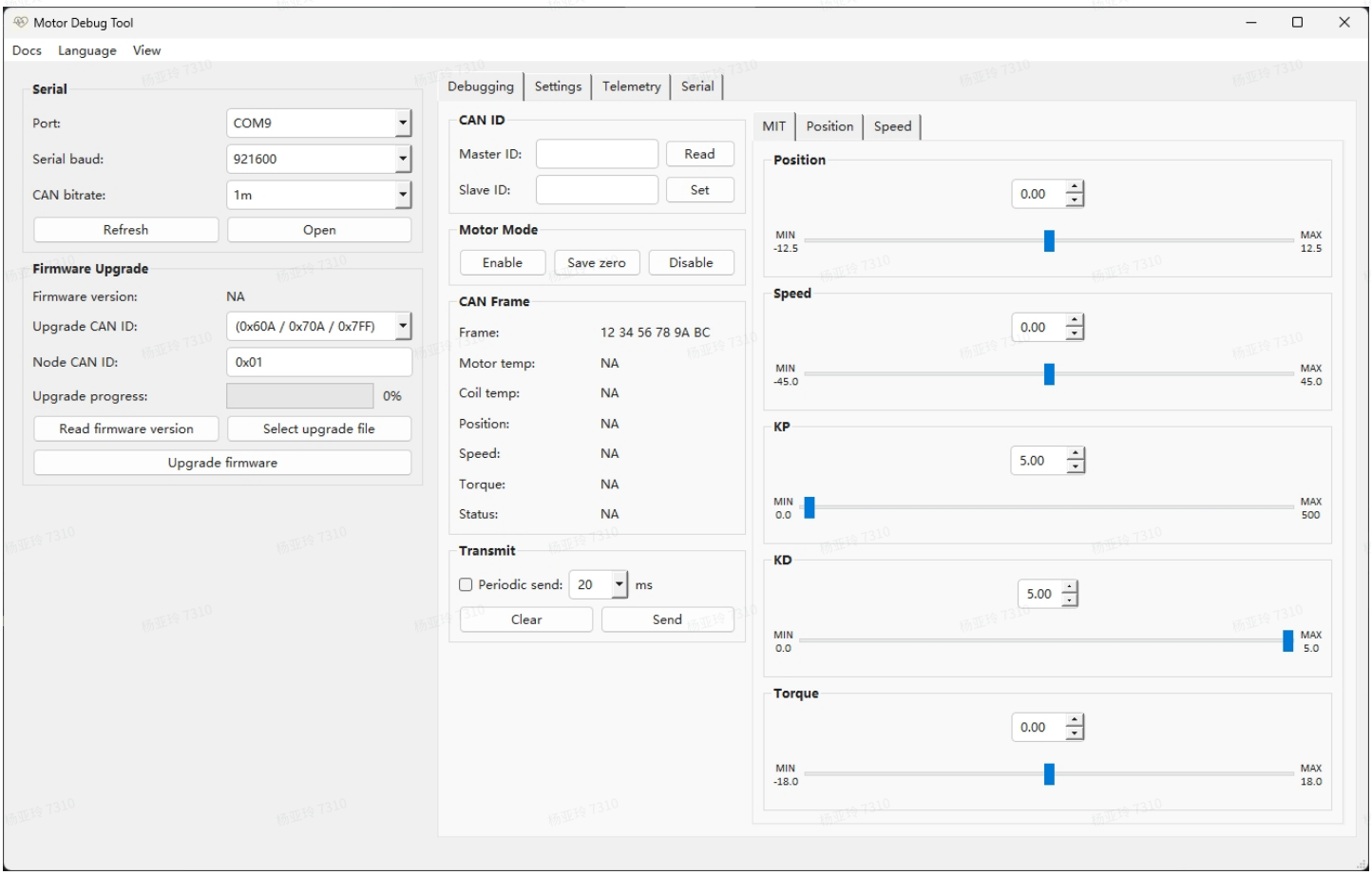
Main Interface Overview

https://github.com/OpenWonderLabs/RobotDoc/blob/main/motor_debug_tool_v1.0.2.zip

1. Click the link above. Download the compressed the file, extract it, and run the .exe file in the folder.
2. Use the Makerbase CANable V2.0 tool to connect to the PC, open the corresponding serial port, and begin functional debugging.

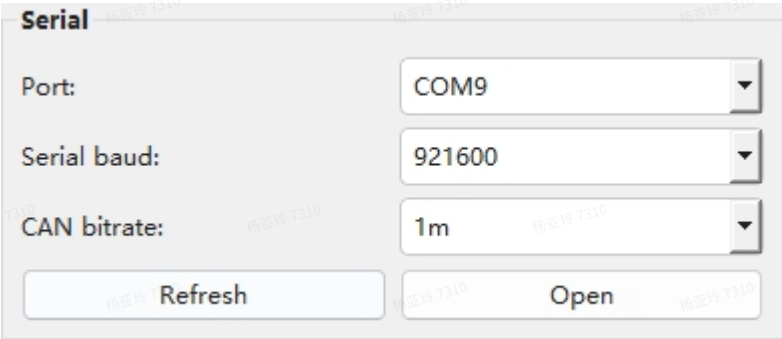
- Note: Before executing any debugging command (except firmware upgrades), you must first set or read the device's CAN ID.

1.1 Menu Bar



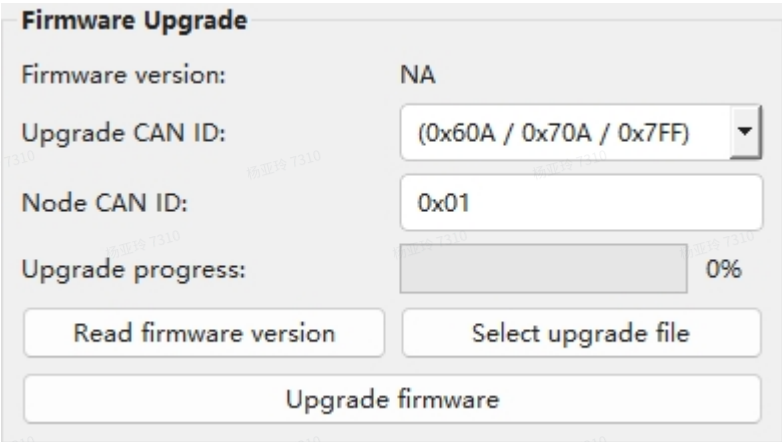
- Operation logs from the host PC software are saved in the `logs` folder.
- Language can be switched between Chinese and English.
- The "Telemetry" page shows data only when you set perotic sending.

1.2 Serial Port Section



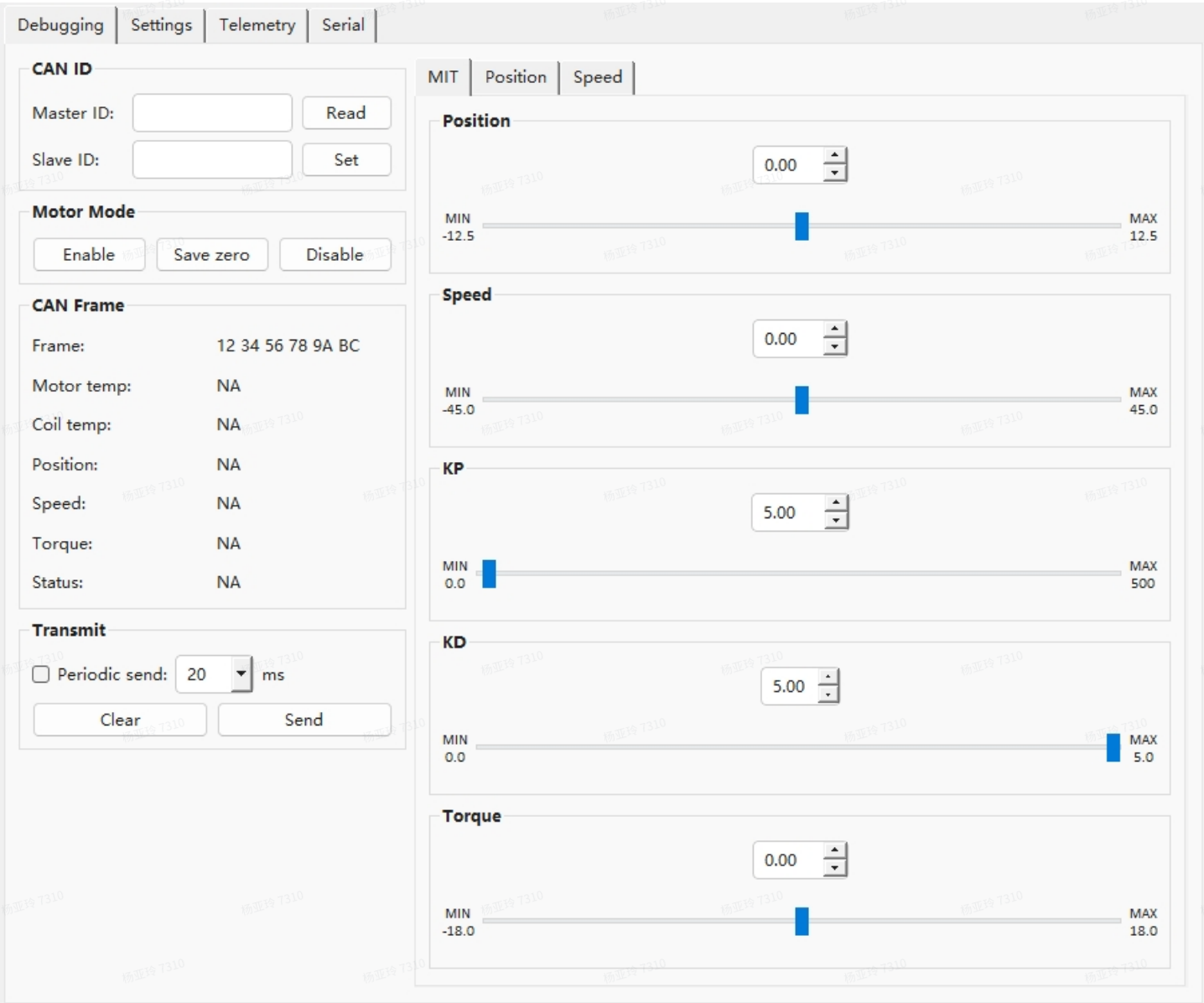
- Supports refreshing, opening, and closing the serial port.
- The current motor only supports 1Mbps CAN baud rate. Use the default configuration.

1.3 Firmware Upgrade Section



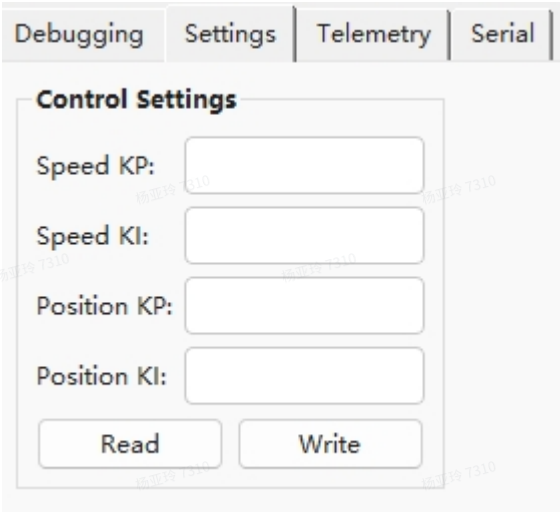
- Supports reading firmware version, selecting a file, and upgrading firmware.
- The upgrade target CAN ID is currently fixed by default.
- The node (device) CAN ID can be entered manually.

1.4 Debugging Page



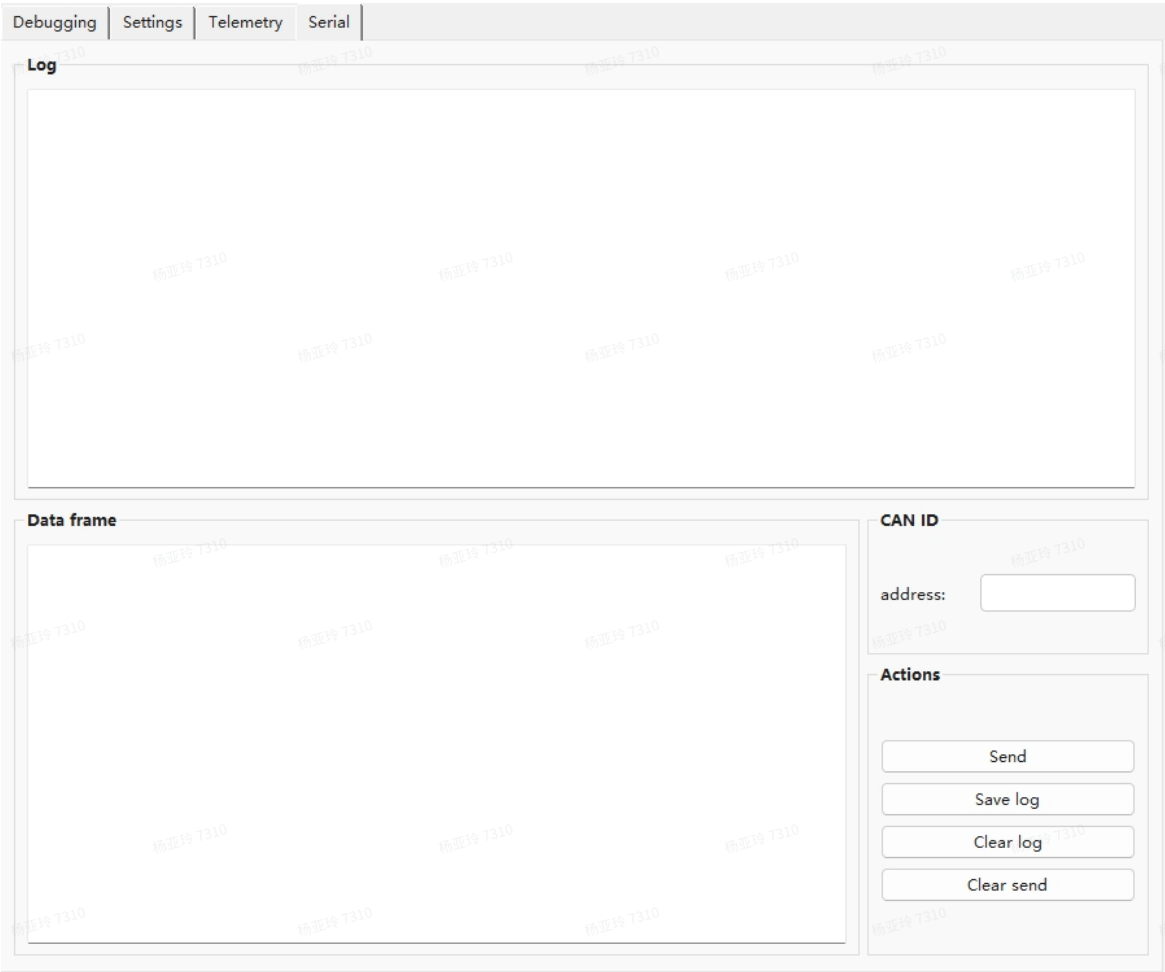
- **CAN ID** section: Supports reading and customizing the ID. Modified CAN IDs take effect after a power cycle.
- **Motor Mode** section: Supports Enable, Save zero, and Disable. Click the corresponding buttons.
- After reading the ID, switch or adjust the control mode (MIT/Position/Speed). The pre-send command is displayed in real-time in the **CAN Frame** section.
- After clicking **Send**, the feedback frame status is parsed and displayed in the **CAN Frame** section.
- **Transmit** section: Set an interval and check the box to send commands continuously. Uncheck to stop.
- When **Periodic send** is selected, data on the **CAN Frame** section and **Telemetry** page updates in real-time.

1.5 Parameter Settings



- Currently only supports reading and setting the parameters as shown above. You can first read the default parameters, then fine-tune and write them back.

1.6 Serial Port Debug



- All commands sent during testing are displayed in the log area.
- Supports manual debugging, requiring you to write the CAN ID and data (in hexadecimal format).